Placebo Trial Design Teacher Notes

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	**Time**	30-40 min
Difficulty	Medium	**ID**	placebo-trial-design

Big idea: Controls protect knowledge from bias.

Overview

Students design a method that protects knowledge from expectation and bias.

Student Prompt

Inspect Placebo Trial Design Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Placebo Trial Design Stimulus A	Student-facing stimulus 1: A mock treatment trial with control, placebo, randomisation, and blinding cards. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Placebo Trial Design Stimulus B	Student-facing stimulus 2: A flawed trial students must repair before trusting its conclusion. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Placebo Trial Design Reveal / Twist	Student-facing stimulus 3: A debrief on ethics when belief itself changes outcomes. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Controls protect knowledge from bias.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Placebo Trial Design Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Placebo Trial Design Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Why is personal experience not enough in science?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how controls protect knowledge from bias.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

• Prepare a starter stimulus using: A mock treatment trial with control, placebo, randomisation, and blinding cards.

• Print or project the student response grid before the activity begins.

• Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

• A mock treatment trial with control, placebo, randomisation, and blinding cards.

• A flawed trial students must repair before trusting its conclusion.

• A debrief on ethics when belief itself changes outcomes.

Suggested Timing

Stage	Teacher move	Watch for
1	Groups design a fair test of the product.	Assumptions, confidence shifts, evidence claims, and language choices.
2	They must include a control group, placebo, blinding, outcome measure, sample size and ethics note.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups exchange designs and find weaknesses.	Assumptions, confidence shifts, evidence claims, and language choices.
4	The class agrees minimum standards for a trustworthy test.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

• Write your first judgement before we discuss it; the first answer is useful evidence.

• Separate what you noticed from what you inferred.

• What would make this knowledge claim more reliable?

Common Misconceptions

• Students may think observation happens before theory in a simple linear order.

• Students may overclaim from a classroom model without naming its limits.

Assessment Look-Fors

• Names variables, assumptions, controls, anomalies, and limits.

• Explains how methods shape what can count as scientific knowledge.

Differentiation and Adaptations

• Short lesson: use only the first example, one response grid, and one debrief question.

• Long lesson: add a second round where students revise the method and compare outcomes.

• Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

• Design a second version of the activity that tests the same knowledge issue in another AOK.

• Find a real-world example where the same knowledge problem matters outside the classroom.

• Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

• Observation is Theory-Laden

• P-hacking with Random Data

• Model Failure Challenge